RPI RENSSELAER POLYTECHNIC INSTITUTE

Vikas Pandey
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Education/ Training

Institution and Location	Degree/Position	Year	Department/ Field
Rensselaer Polytechnic Institute Troy, New York, USA	Research Scientist	2024 - present	Biomedical Optics, Machine Learning, and Cancer Imaging
Rensselaer Polytechnic Institute Troy, New York, USA	Postdoctoral Research Associate	2022 - 2024	Biomedical Optics, Machine Learning, and Cancer Imaging
Valetude Primus Healthcare New Delhi, India	Co-founder, Managing Director	2015 - 2021	Infectious Diseases, Biomedical Optics, Healthcare Devices, Clinical Validations
Indian Institute of Technology New Delhi, India	Ph.D.	2014 - 2020	Chemical Engineering
Evalueserve Pvt. Ltd	Business Analyst	2011 - 2013	Data Modeling, Market Analysis
Indian Institute of Technology New Delhi India	Bachelor and Master of Technology (Dual Degree)	2006 - 2011	Biochemical Engineering and Biotechnology

Patents

- A composition for mucus or sputum liquefaction and a process thereof **(Indian Patent no. – 361905)**
- Compact evanescent wave-based fluorescence illumination **(Indian Patent no. – 382141)**

Publications

Year	Journal	Impact Factor	Detail
2026	ICLR 2026	**~**	*Compositional amortized inference for large-scale hierarchical Bayesian models;* Jonas Arruda, **Vikas Pandey**, Catherine Sherry, Margarida Barroso, Xavier Intes, Jan Hasenauer, Stefan T. Radev
2026	Journal of Biophotonics	**2.3**	*Enhancing Fluorescence Lifetime Imaging with a Differential transformer;* Ismail Erbas, **Vikas Pandey**, Navid Ibtehaj Nizam, Nanxue Yuan, Amit Verma, Tynan Young, John Williams, Margarida Barosso, Xavier Intes,
2025	PhotoniX (Springer Nature)	**19.1**	*Real-Time Wide-field Fluorescence Lifetime Imaging via Single-Snapshot Acquisition for Biomedical Applications*; **Vikas Pandey***, Euan Millar*, Ismail Erbas, Luis Chavez, Jack Radford, Isaiah Crosbourne, Mansa Madhusudan, Gregor G. Taylor, Nanxue Yuan, Claudio Bruschini, Stefan Radev, Margarida Barroso, Andrew Tobin, Xavier Michalet, Edoardo Charbon, Daniele Faccio, Xavier Intes
2025	Journal of Physics: Photonics	**8.4**	*Isolating subsurface fluorescence in macroscopic lifetime imaging via high-spatial-frequency structured illumination;* Nanxue Yuan, Saif Ragab, Navid Nizam, **Vikas Pandey**, Amit Verma, Tynan Young, John Williams, Margarida Barroso, Xavier Intes

2025	NeurIPS 2025 @ Imageomics Workshop	~	*EvidenceMoE: A Physics-Guided Mixture-of-Experts with Evidential Critics for Advancing Fluorescence Light Detection and Ranging in Scattering Media;* Ismail Erbas, Ferhat Demirkiran, Karthik Swaminathan, Naigang Wang, Navid Ibtehaj Nizam, Stefan T. Radev, Kaoutar El Maghraoui, Xavier Intes, **Vikas Pandey**
2025	Optics Letters	**3.3**	*Evaluating Tartrazine as an Optical Clearing Agent for Fluorescence Lifetime Imaging;* Nanxue Yuan, Saif Ragab, Luis Chavez, **Vikas Pandey**, Xavier Intes
2025	Biomedical Optics Express	**3.2**	*Design and Characterization of an Optical Phantom for Mesoscopic Multimodal Fluorescence Lifetime Imaging and Optical Coherence Elastography;* Luis Chaves, Shan Gao, **Vikas Pandey**, Nanxue Yuan, Saif Ragab, Jiayue Li, David Corr, Brendan Kennedy, Xavier Intes
2024	Advanced Science	**14.1**	*Fluorescence Lifetime Imaging for Quantification of Targeted Drug Delivery in Varying Tumor Microenvironments*; Amit Verma*, **Vikas Pandey***, Catherine Sherry, Christopher James, Kailie Matteson, Jason T. Smith, Alena Rudkouskaya, Xavier Intes, Margarida Barroso
2024	Optics Letters	**3.3**	*Deep Learning-based Temporal Deconvolution for Photon Time-of-Flight Distribution Retrieval*; **Vikas Pandey**, Ismail Erbas, Xavier Michalet, Arin Ulku, Claudio Bruschini, Edoardo Charbon, Margarida Barroso, Xavier Intes
2024	NeurIPS 2024 @ Compression Workshop	~	*Compressing Recurrent Neural Networks for FPGA-accelerated Implementation in Fluorescence Lifetime Imaging*; Machine Learning and Compression Workshop; Ismail Erbas*, **Vikas Pandey***, Aporva Amarnath, Naigang Wang, Karthik Swaminathan, Stefan T. Radev, Xavier Intes
2022	Particle & Particle Systems Characterization	**3.0**	*SNC-TIRS: Label-Free Single Nanoparticle Characterization System Based on Total Internal Reflection Scattering Signal;* Arti Tyagi, Shefali Singh, **Vikas Pandey**, Sachin Kumar, Prabhat Singh Mallik, Ravikrishnan Elangovan
2019	Scientific reports	**3.9**	*SeeTB: A novel alternative to sputum smear microscopy to diagnose tuberculosis in high burden countries* **Vikas Pandey***, Pooja Singh*, Saumya Singh, Naresh Arora, Neha Quadir, Saurabh Singh, Ayan Das, Mridu Dudeja, Prem Kapur, Nasreen Zafar Ehtesham, Ravikrishnan Elangovan, Seyed E Hasnain
2017	Methods and Applications in Fluorescence	**2.4**	*Compact 3D printed module for fluorescence and label-free imaging using evanescent excitation* **Vikas Pandey**, Shalini Gupta, Ravikrishnan Elangovan
2016	Advanced Materials Technologies	**6.8**	*Spot immunomagnetic enrichment device for rapid detection of pathogens in peripheral blood* Saurabh Singh, Mohita Upadhyay, **Vikas Pandey**, Perumal Vivekanandan, Shalini Gupta, Ravikrishnan Elangovan
2016	Scientific Reports	**3.9**	*Maximum limit to the number of myosin II motors participating in processive sliding of actin* Khushboo Rastogi, Mohammed Shabeel Puliyakodan, **Vikas Pandey**, Sunil Nath, Ravikrishnan Elangovan

*** Shared First Authors**

Archive/ preprint manuscripts

Year	Status	Detail

2025	arXiv	*Quantifying Lifetime Uncertainty in Gated-ICCD Fluorescence Lifetime Imaging*; Nanxue Yuan, **Vikas Pandey**, Amit Verma, Taylor Humphrey, Margarida Barroso, Xavier Intes, and Xavier Michalet
2024	arXiv:2410.07364	*Unlocking Real-Time Fluorescence Lifetime Imaging: Multi-Pixel Parallelism for FPGA-Accelerated Processing*; Ismail Erbas, Aporva Amarnath, **Vikas Pandey**, Karthik Swaminathan, Naigang Wang, Xavier Intes
2024	Research Square	*Transformer-based Deep Learning Model for Fluorescence Lifetime Parameter Estimations using Pixelwise Instrument Response Function*; Ismail Erbas, **Vikas Pandey**, Navid Ibtehaj Nizam, Nanxue Yuan, Amit Verma, Margarida Barroso, Xavier Intes
2024	bioRxiv	*A Novel Technique for Fluorescence Lifetime Tomography;* Navid Ibtehaj Nizam, **Vikas Pandey**, Ismail Erbas, Jason T Smith, Xavier Intes

Book/ Book Chapter

Year	Book	Editors	Detail
2024	"Cancer Detection and Diagnosis."	Rasooly, A., Prickril, B., Ossandon, M.R.	*"Near-infrared macroscopic and mesoscopic fluorescence lifetime FRET imaging to measure intra-tumor heterogeneity of antibody-target engagement,"* (book chapter) https://doi.org/10.1201/9781003449942 A Verma, S Ragab, S Gao, C Sherry, N Yuan, **V Pandey**, X Intes, M Barroso,

Conference Presentations/Papers

Year	Conference	Detail
2026	Molecular-Guided Surgery: Molecules, Devices, and Applications	*Invited Presentation – Intelligent Widefield Fluorescence Lifetime Imaging Across Scales* XR Intes (**Vikas Pandey** - presenter)
2026	SPIE Photonics West 2026	*Real-time Fluorescence Lifetime Imaging for Multi-scale Biomedical Applications* **Vikas Pandey***, Euan Millar*, Ismail Erbas, Luis Chavez, Jack Radford, Isaiah Crosbourne, Mansa Madhusudan, Gregor G. Taylor, Nanxue Yuan, Claudio Bruschini, Stefan Radev, Margarida Barroso, Andrew Tobin, Xavier Michalet, Edoardo Charbon, Daniele Faccio, Xavier Intes
2025	International Symposium on Fluorescence Guided Surgery **MEEI Harvard Medical School**	*Invited talk - AI-Driven Fast Widefield Fluorescence Lifetime Imaging for Surgical Guidance (Mike Toth Research Center, Mass Eye and Ear)* **Vikas Pandey**
2025	Molecular-Guided Surgery: Molecules, Devices, and Applications XI, PC1330106	*Invited Presentation - AI-enhanced widefield fluorescence lifetime imaging at multi-scale* XR Intes (**Vikas Pandey** - presenter)
2025	ACM/SIGDA International Symposium on FPGA 2025	*No Time to Lose: Enabling Real-Time Fluorescence Lifetime Imaging on Resource-constrained FPGAs Through Efficient Scheduling;* Ismail Erbas, Aporva Amarnath, **Vikas Pandey**, Karthik Swaminathan, Naigang Wang, Xavier Intes
2025	SPIE Photonics West 2025	*AI-enhanced rapid lifetime determination method for fast macroscopic and mesoscopic fluorescence lifetime imaging* **Vikas Pandey**, Ismail Erbas, Luis Chavez, Claudio Bruschini, Edoardo Charbon, Margarida Barroso, Stefan Radev, Xavier Michalet, Xavier Intes

2025	SPIE Photonics West 2025	*Integrating time-resolved NIR and SWIR imaging for high-resolution mesoscopic fluorescence lifetime imaging using deep learning* **Vikas Pandey**, Stefan Radev, Luis Chavez, Ismail Erbas, Claudio Bruschini, Edoardo Charbon, Xavier Intes
2025	SPIE Photonics West 2025	*Mesoscopic light-sheet imaging set-up for 3D SWIR fluorescence intensity and NIR fluorescence lifetime imaging* Luis Chavez, Ismail Erbas, **Vikas Pandey**, Catherine Sherry, Isaiah Crosbourne, Claudio Bruschini, Edoardo Charbon, Margarida Barroso, Xavier Intes
2025	SPIE Photonics West 2025	*Confidence in NLSF-based lifetime estimation through LUT in time-gated ICCD-based macroscopic fluorescence lifetime imaging* Nanxue Yuan, **Vikas Pandey**, Xavier Intes, Xavier Michalet
2025	SPIE Photonics West 2025	*Design and characterization of a single-pixel light sheet mesoscope for fluorescence imaging* Navid Ibtehaj Nizam, Luis Chavez, **Vikas Pandey**, Ismail Erbas, Xavier Intes
2025	SPIE Photonics West 2025	*Characterization and validation of a phantom design for multimodal mesoscopic fluorescence lifetime and optical coherence elastography* Luis Chavez, Shan Gao, **Vikas Pandey**, Nanxue Yuan, Saif Ragab, Jiayue Li, Matt S. Hepburn, Percy Smith, David T. Corr, Brendan F. Kennedy, Xavier Intes
2025	SPIE Photonics West 2025	*FPGA implementation of sequence-to-sequence encoder-decoder deep learning model for real-time fluorescence parameter estimation through SwissSPAD2 camera* Ismail Erbas, Paul Mos, **Vikas Pandey**, Claudio Bruschini, Edoardo Charbon, Xavier Intes
2025	SPIE Photonics West 2025	*Antibody-to-imaging pipeline to monitor target engagement in breast tumors* Margarida M. Barroso, Amit Verma, Cay Sherry, Nanxue Yuan, **Vikas Pandey**, Tynan Young, John Williams, Xavier Intes
2024	NeurIPS 2024	*Compressing Recurrent Neural Networks for FPGA-accelerated Implementation in Fluorescence Lifetime Imaging*; Machine Learning and Compression Workshop Ismail Erbas, **Vikas Pandey**, Aporva Amarnath, Naigang Wang, Karthik Swaminathan, Stefan T. Radev, Xavier Intes
2024	IBM/IEEE AI Compute Symposium 2024 (AICS'24)	*A Leap Forward in Cancer Imaging: Optimizing Recurrent Neural Networks for Real-Time Fluorescence Lifetime Imaging on Resource-Constrained FPGAs* Ismail Erbas, **Vikas Pandey**, Aporva Amarnath, Naigang Wang, Karthik Swaminathan, Xavier Intes
2024	Imagine AI 2024 (Singapore)	*AI-Enhanced Rapid Fluorescence Lifetime Estimation with SwissSPAD3 for Fluorescence-Guided Cancer Surgery;* **Vikas Pandey**, Ismail Erbas, Claudio Bruschini, Edoardo Charbon, Margarida Barroso, Stefan T. Radev, Xavier Michalet, Xavier Intes
2024	Imagine AI 2024 (Singapore)	*Enabling Real-Time Fluorescence Lifetime Imaging on FPGAs through Model Compression and Efficient Scheduling;* Ismail Erbas*, **Vikas Pandey***, Aporva Amarnath, Karthik Swaminathan, Naigang Wang, Stefan T. Radev, Xavier Intes
2024	Optica Biophotonics Congress 2024	*Temporal Point Spread Function Deconvolution in Time-resolved Fluorescence Lifetime Imaging using Deep Learning Model;* **V Pandey**, I Erbas, X Michalet, A Ulku, C Bruschini, E Charbon, X Intes
2024	Optica Biophotonics Congress 2024	*Deep Learning Aided Fluorescence Lifetime Tomography;* NI Nizam, I Erbas, **V Pandey**, JT Smith, X Intes

2024	Optica Biophotonics Congress 2024	*Fluorescence lifetime parameters estimation with transformer based deep learning model;* I Erbas, **V Pandey**, NI Nizam, N Yuan, X Intes
2024	Optica Biophotonics Congress 2024	*Experimental study of Fluorescence Lifetime Uncertainty in Time-Gated ICCD-based Macroscopic Fluorescence Lifetime Imaging* N Yuan, **V Pandey**, X Michalet, X Intes
2024	Optica Biophotonics Congress 2024	*Multimodal Fluorescence Lifetime Imaging and Optical Coherence Elastography for Mesoscopic Structural, Biomechanical, and Molecular Imaging;* Luis Chavez, Shan Gao, **Vikas Pandey**, Nanxue Yuan, Jiayue Li, Matt S Hepburn, Percy Smith, Caroline Edelheit, David T Corr, Brendan F Kennedy, Xavier Intes
2024	Optica Biophotonics Congress 2024	*Monte-Carlo Based Data Generator for Fluorescence Lifetime Applications* NI Nizam, I Erbas, **V Pandey**, X Intes
2024	Optica Biophotonics Congress 2024	*Multimodal Data-fusion in Fluorescence Lifetime Imaging using Deep Learning* **V Pandey**, I Erbas, X Intes
2024	Optica Biophotonics Congress 2024	*Quantifying Drug-Receptor Engagement using Macroscopic Fluorescence Lifetime FRET in vivo Imaging* A Verma, **V Pandey**, N Yuan, C Sherry, T Humphrey, C James, Tynan Young, John C Williams, Xavier Intes, Margarida Barroso
2024	SPIE Photonics West 2024	*Deep Learning model for efficient instrument response function deconvolution in fluorescence lifetime imaging* **V Pandey**, X Michalet, A Ulku, C Bruschini, E Charbon, X Intes
2024	SPIE Photonics West 2024	*Using meditope-based antibody labeling to improve fluorescence lifetime FRET imaging* A Verma, C Sherry, N Yuan, **V Pandey**, J Williams, X Intes, MM Barroso
2024	SPIE Photonics West 2024	*Fluorescence lifetime tomography via deep learning* NI Nizam, **V Pandey**, JT Smith, I Erbas, X Intes
2024	SPIE Photonics West 2024	*3D fluorescence molecular tomography utilizing a novel SPAD camera* NI Nizam, **V Pandey**, I Erbas, A Ulku, C Bruschini, E Charbon, Xavier Michalet, Xavier Intes
2024	SPIE Photonics West 2024	*Transformer based deep learning model for fluorescence lifetime parameters estimation using pixelwise instrument response function* I Erbas, **V Pandey**, NI Nizam, N Yuan, X Intes
2024	SPIE Photonics West 2024	*Macroscopy fluorescence lifetime FRET monitors drug-target engagement in tumors in vivo* A Verma, C Sherry, **V Pandey**, N Yuan, J Williams, X Intes, MM Barroso
2024	SPIE Photonics West 2024	*Antibody-target binding quantification in living tumors using macroscopy fluorescence lifetime Forster resonance energy transfer imaging (MFLI FRET)* N Yuan, **V Pandey**, A Verma, JC Williams, X Intes, M Barroso
2023	SPIE Photonics West 2023	*Impact of tumor microenvironment in antibody-HER2 binding using macroscopy fluorescence lifetime FRET in vivo imaging* A Verma, **V Pandey**, JT Smith, X Intes, M Barroso
2023	SPIE Photonics West 2023	*Near infrared fluorescence lifetime FRET microscopy to evaluate antibody drug binding in various HER2 positive cancer cell lines* Catherine Sherry, Amit Verma, Jason Smith, **Vikas Pandey**, Tynan Young, John Williams, Xavier Michalet, Xavier Intes, Margarida Barroso
2019	4th WHO Global Forum on Medical Devices	*A novel portable device and thinning reagent for highly sensitive TB detection at low resource settings:* **Vikas Pandey**, Seyed E Hasnain & Ravikrishnan Elangovan

2019	5th Global Forum on TB Vaccines	*Miniaturized Fluorescence adapter for Fluorescence Sputum Smear Microscopy using bright-field microscope;* **Vikas Pandey**, Pooja Singh, Mamta Rani, Saumya Singh, Nasreen Z Ehtesham, Seyed E Hasnain & Ravikrishnan Elangovan
2017	Biophysical Journal	*Minimum and Maximum Limit to Number of Myosin II Motors Participating in an Ensemble Motility*; Khushboo Rastogi, **Vikas Pandey**, Sunil Nath, Ravikrishnan Elangovan
2016	OWLS FCS	*Compact waveguide-based large area evanescent excitation for label-free and fluorescent imaging of very small bio-molecules;* **Vikas Pandey**, Shalini Gupta & Ravikrishnan Elangovan

Research Experience

Research Scientist (Aug 2024 – current)

Centre for Modeling, Simulation & Imaging in Medicine (CeMSIM) & Biomedical Engineering
Rensselaer Polytechnic Institute, Troy, NY

- *In vivo* non-invasive optical imaging methods for drug internalization quantification in HER2+ tumor xenografts in live small animals (mice) through FLI.
- Developing novel algorithms (deep learning and AI hardware accelerators) for real-time Fluorescence Lifetime Imaging (FLI), for translation to image guided surgery.
- Developing macro- and mesoscopic optical imaging set-up for FLI; developing single-pixel imaging and light-sheet based mesoscopic imaging setups.
- Developing FLI data processing methods using Deep Learning models, and Hierarchical Bayesian inferences.
- Co-Principal Investigator (Co-PI) in the RPI-IBM Artificial Intelligence Research Collaboration (AIRC), focusing on neural network-accelerated hardware development for FLI.
- Forging new collaborations and nurturing existing partnerships through joint research projects and proposals.
- Mentoring 1 PhD student and guiding 2 undergraduate students on their research projects.

Post-doctoral Research Associate (Jan 2022 – Aug 2024)

Biomedical Engineering & CeMSIM, Rensselaer Polytechnic Institute, Troy, NY

- *In vivo* non-invasive optical imaging method for drug internalization quantification in HER2+ tumour xenografts in live intact small animals (mice) through Fluorescence Lifetime Imaging (FLI).
- Machine learning model development to simplify and fast analysis of the complex ill-defined mathematical inverse problem

ERASMUS + KA107 Ph.D. (Research exchange between India & Spain)

Optics Department, Universidad de Santiago de Compostela, Spain

- Selected from the Chemical Engineering Department IIT Delhi for research exchange in Spain.
- Focused on waveguide development for Total Internal Reflection Fluorescence (TIRF) scattering imaging.

Co-founder, Managing Director (Jun 2015 – August 2021)

Valetude Primus Healthcare Pvt. Ltd.

- Led multiple healthcare technologies development as Principal Investigator (PI), developing prototypes and full products; worked with investors and government agencies for securing funds and business development.
- Conducted multi-centeric clinical validations and received certification for TB-product from the Drug Controller General of India.

Judgement/Guest Lecture/Expert in Panel

- **Session Chair (Deep Learning):** SPIE Photonics West 2026, Multimodal Biomedical Imaging XIX, Conference, January, 2026, San Francisco, CA, USA
- **Participation:** Symposium on Optical Surgical Navigation 2026**,** Stanford Medicine, Palo Alto, Stanford, USA.

- **Invited Presentation:** SPIE Photonics West 2025, Invited presentation (with Prof. Xavier Intes) on Molecular-Guided Surgery: Molecules, Devices session.
- **Session Chair (Deep Learning):** SPIE Photonics West 2024, Multimodal Biomedical Imaging XIX, Conference 12834, January 27, 2024, San Francisco, CA, USA.
- **Podium Presentation -National Cancer Institute (NCI/NIH), USA:** Junior Investigator podium presentation with financial coverage, July 2024
- **Invited Presentation:** Webinar - Deep learning techniques enhance fluorescence imaging by Electro Optics & Hamamtsu Photonics, 2023.
- **Research Judge:** Research Symposium – Judged Poster Sessions for the Centre for Materials, Devices, and Integrated Systems (CMDIS), Rensselaer Polytechnic Institute, Troy, New York in 2023 & 2022.
- **Invited Participant:** Geneva Health Innovation Fair for innovation in tuberculosis (TB) diagnostics technology.
- **Invited Guest Lecture:** Grassroots innovation at the top Indian Management School, Indian Institute of Management (IIM) Ahmedabad, Gujarat, India, 2019.
- **Invited Panelist:** Intel Research - Discussed the role of Artificial Intelligence in Healthcare at Intel Research, Bengaluru, India, 2018.
- **Panelist (BIRAC):** Addressed infectious disease innovative technologies at the Biotechnology Industry Research Assistance Council (BIRAC), New Delhi, India, 2018.
- **Panelist (Invest India):** Discussed strengthening the innovation ecosystem for biopharmaceuticals at an Invest India, Government of India meeting, 2018.
- **Panelist (BIRAC):** Addressed antimicrobial resistance at the Biotechnology Industry Research Assistance Council (BIRAC), Government of India, 2017.
- **Panelist (IIT):** Spoke on innovation in healthcare at the Indian Institute of Technology (IIT) Delhi, 2017.

Honors

- **Membership:** *Sigma Xi*, the scientific research honour society since 2024
- **Lifetime Member (Optics & Photonics societies, USA):** Optica (formerly OSA) www.optica.org , and SPIE (https://www.spie.org/)
- **Invited Member:** Sepsis Innovation Collaborative, California, USA, 2022.
- **Central Drugs Standard Control Organisation (CDSCO):** Certificate from the Drug Controller General of India for the SeeTB device implementation in resource-poor settings, 2020.
- **Gandhian Young Technology Innovation Award:** Conferred by the honourable President of India for the development of an effective tuberculosis diagnostics method in resource-limited settings, 2018.
- **Social Alpha Impact Innovation Award**, **Grand Challenge Exploration Award** from BIRAC, Bill & Melinda Gates Foundation, and IKP, **Biotechnology Ignition Grant (BIG)** from BIRAC, 2018.
- **Longitude Prize Discovery Award: Royal** Society London for innovation in Antimicrobial Resistance (AMR) detection, 2016.
- **Elected House Secretary**: Represented 800 students of Udaigiri House at IIT Delhi in 2015, addressing their internal and external affairs and advocating their concerns with the IIT Delhi administration.

Grants and Awards

Name		Type	Year	Amount
IBM-RPI Future of Computing Research Collaboration (FCRC) (**Co-PI**)	USA	Grant	2025	**100,000 USD**
IBM-RPI AI Research collaboration (AIRC) (**Co-I**)	USA	Grant	2024	**180,000 USD**
Biotechnology Ignition Grant (**PI**)	India	Grant	2019	**5,000,000 INR**
India Health Fund (**Co-I**)	India	Grant	2019	**19,000,000 INR**
President Award (Gandhian Young Technology Innovation Award) (**PI**)	India	Award	2018	**1,500,000 INR**

Grand Challenge Exploration (Bill & Melinda Gates Foundation & BIRAC) (**PI**)	India	Grant	2018	**5,000,000 INR**
Social Alpha – Quest for Innovation (**PI**)	India	Grant	2018	**2,500,000 INR**
Tech Planter (**Co-PI**)	Japan	Award	2016	**200,000 YEN**
Longitude Prize Discovery Award (**PI**)	UK	Grant	2016	**1,150,000 INR**
Pfizer – IIT Delhi Innovation Award (**Co-PI**)	India	Grant	2016	**5,000,000 INR**
IIT Delhi Innovation Award & Seed Funding (**Co-PI**)	India	Award	2015	**200,000 INR**

INR – Indian Currency; **YEN** – Japanese Currency; **USD** – USA Currency

Teaching and Workshops

- **Co-Instructor:** Department of Biomedical Engineering, Rensselaer Polytechnic Institute, co-teaching with Prof. Xavier Intes
    - Biophotonics: Spring 2023, Spring 2024
    - Bio-Imaging and Bio-Instrumentation: Fall 2022, Fall 2023, Fall 2024
- **Workshop Attendee:** American Institute for Medical and Biological Engineering (AIMBE) Public Policy Institute, Washington, DC, October 2023
    - Focused on the role of researchers in public policy and advocacy.
- **Participant:** SwissSPAD User Group Meeting and Workshop by Advanced Quantum Architecture Laboratory (AQUA), École Polytechnique Fédérale de Lausanne (EPFL), Switzerland
    - Creativity Meeting: March 2023, Les Diablerets, Switzerland
    - User Group Meeting: May 2024, Les Diablerets, Switzerland
- **Participant:** AI Hardware Forum (Invite Only) Nov 2024, at IBM Research at Thomas J. Watson Research Centre, Yorktown Heights, NY, USA
- **Organizer:** Antimicrobial Resistance (AMR) Workshop with Prof. Ravikrishnan Elangovan, Indian Institute of Technology (IIT) Delhi, India, November 2018

Scientific Journal & Grant Reviews

Editor for scientific journals	
2025	1. Guest Associate Editor Frontiers in Medicine (Precision Medicine); **Topic:** Next-Generation Preclinical Imaging and Analytical Technologies for Personalized Oncology 2. Guest Associate Editor Frontiers in Medicine (Brain Imaging Methods); **Topic:** Advancing neuroimaging diagnostics with machine learning and computational models
Reviewer for scientific journals	
Year	**Journals**
2025	Frontiers in Artificial Intelligence - AI in Food, Agriculture and Water Frontiers in Artificial Intelligence Brain Imaging Methods
2024	Nature Communications Engineering IEEE Transactions on Biomedical Engineering Journal of Biomedical Optics Biomedical Optics Express
2023	Nature Communications IEEE Transactions on Biomedical Engineering Optica (Optics Letters) Optica (Applied Optics)

	Biomedical Optics Express Journal of Biomedical Optics
2022	Journal of Biomedical Optics Frontier Sciences
Reviewer for grants	
Year	**Grants/Awards**
2022	Co-reviewed grant with Prof. Xavier Intes for Swiss National Science Foundation
2020	Gandhian Young Technology Innovation Award (SRISTI)
2019	Gandhian Young Technology Innovation Award (SRISTI)

Industry Experience

Board member **(Jan 2022 – 2024)**
Valetude Primus Healthcare Pvt Ltd
- Participate in strategic financial and operational planning for the company.
- Provide scientific expertise in technology and product development.

Co-founder, Managing Director **(May 2015 – Dec 2021)**
Valetude Primus Healthcare Pvt Ltd
- Established the company, building a team of young professionals, aligning business objectives with the company's mission, and developing patented technologies.
- Collaborated on integrated strategies to drive business growth, managing relationships with stakeholders, including investors, manufacturers, customers, regulators, and academics.
- Developed an executive team focused on achieving ambitious targets with well-structured plans.
- Led strategic initiatives to enhance operational efficiency and streamline organizational processes.
- Set and optimized internal policies to ensure responsiveness and scalability.
- Secured patents in the US, UK, South Africa, and India for four innovative technologies developed for commercialization.

Business Analyst **(June 2011 – Dec 2012)**
Evalueserve Pvt Ltd
- Developed statistical models for forecasting and prediction in data science projects.
- Performed competitive benchmarking to identify cost-saving opportunities and potential product enhancements.
- Analyzed data from primary and secondary sources to inform business strategy and optimize workflows.

Undergrad Internship
- *AstraZeneca India*, Bangalore, Karnataka **(May 2009 – August 2009)**
- *Wake Forest University*, Winston-Salem, North Carolina, USA **(May 2008 – August 2008)**

Extra-Curricular Activities
- **Coding**: Enjoy coding as a hobby to solve problems and explore new technologies.
- **Traveling**: Passionate about exploring new places, especially beach destinations.
- **Reading**: Avid reader of articles across various domains, staying updated on new trends and insights.
- **Games**: Enthusiastic about playing pool and chess, which help sharpen strategic thinking.